Ex: 9 Implementation of subnetting in CISCO PACKET TRACER Simulator

AIM:

To implement classless IP subnetting and test network connectivity using CISCO packet tracer.

PROCEDURE:

Open CISCO packet tracer and create a new network topology.

Add the required routers, switches and PC's to the workspace.

Connect the devices using appropriate cables.

Dividing the network 192.168.1.0/24 into smaller subnets using a /27 subnet mask

Assign suitable IP addresses and subnet masks to the router interfaces.

Configure the PC's with their respective IP address subnet mask & default gateway.

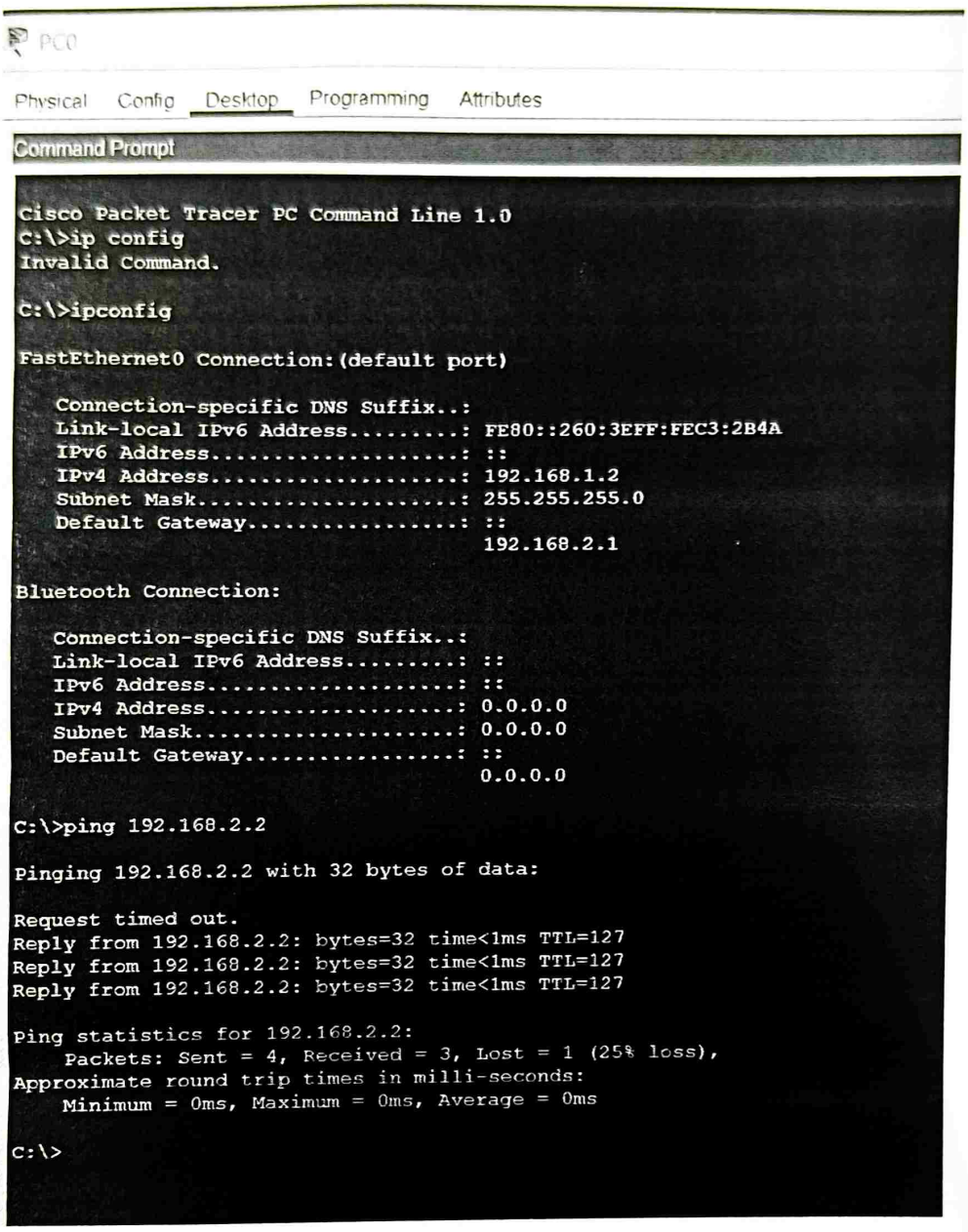
Enable the router interfaces using the no shutdown command.

Using ping command from the PC's to test connectivity.

Verify that the packets are successfully Transmitted b/t the interfaces.

CLI Command:

```
# enable
# Configure Terminal
# interface Fast Ethernet 0/0
# ip address 198.168.1.10 256.255.255.0
# no shutdown
# exit
```



Subnet 1 to Subnet 2 communication extablishment: Similarly for all the remaining 2 subnets and the remaining 2 PC's

```
Router#show startup-conf
Using 732 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX1524WFBP-
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.2.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 ip address 192.168.3.1 255.255.255.0
 duplex auto
 speed auto
!
interface Vlan1
 no ip address
 shutdown
!

Router#
```

Tracer.

Result:

Thus, classless IP subnetting was successfully implemented and tested using CISCO packet Tracer.